

Course Name:	Digital Signal and Image Processing	Semester:	VI
Date of Performance:	13/ 02/ 2026	DIV/ Batch No:	C-3
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Title: Implement the following point processing techniques in spatial domain:

- Image Negative.
- Thresholding.
- Gray level slicing with and without background
- Bit plane slicing
- Contrast stretching

Objective: To learn & understand point processing techniques.

Expected Outcome of Experiment:

CO	Outcome
CO4	Apply various spatial and frequency domain enhancement techniques for 1-D and 2-D signals.

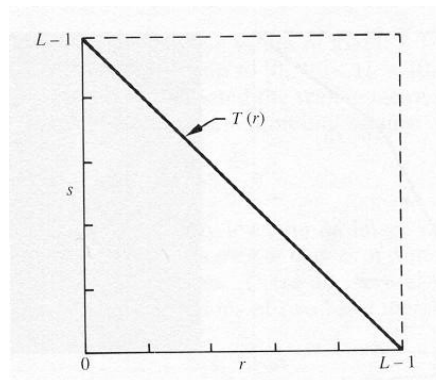
Books/ Journals/ Websites referred:

1. <http://www.mathworks.com/support/>
2. www.math.mtu.edu/~msgocken/intro/intro.html.
3. R. C.Gonsales R.E.Woods, "Digital Image Processing", Second edition, Pearson Education
4. S.Jayaraman, S Esakkirajan, T Veerakumar "Digital Image Processing "Mc Graw Hill.
5. S.Sridhar,"Digital Image processing", oxford university press, 1st edition."

Pre Lab/ Prior Concepts:

Image Negative:

Negative images are useful for enhancing white or grey detail embedded in dark regions of an image. Image negatives are obtained by using the transformation function $s=T(r)$.

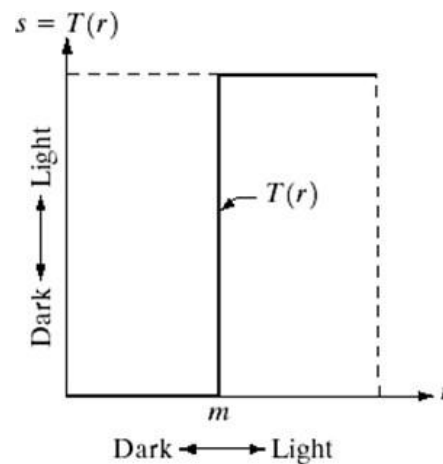


$[0, L-1]$ is the range of gray levels

$$S = L-1-r$$

Thresholding

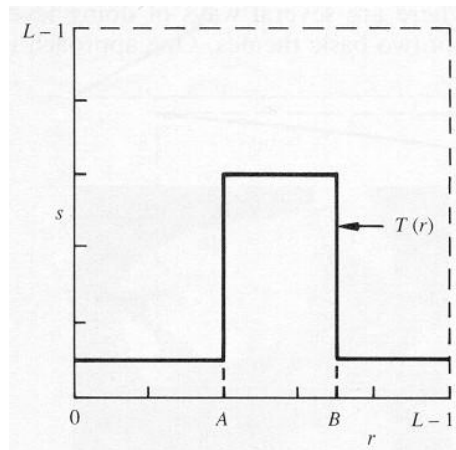
From a grayscale image, thresholding can be used to create binary images. The simplest thresholding methods replace each pixel in an image with a black pixel if the image intensity is less than some fixed constant T or a white pixel if the image intensity is



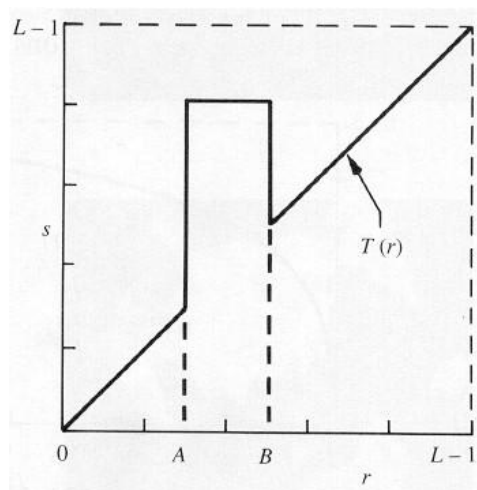
greater than that constant.

Gray Level Slicing

To highlight a specific range of gray levels in an image (e.g. to enhance certain features). One way is to display a high value for all gray levels in the range of interest and a low value for all other gray levels (binary image).

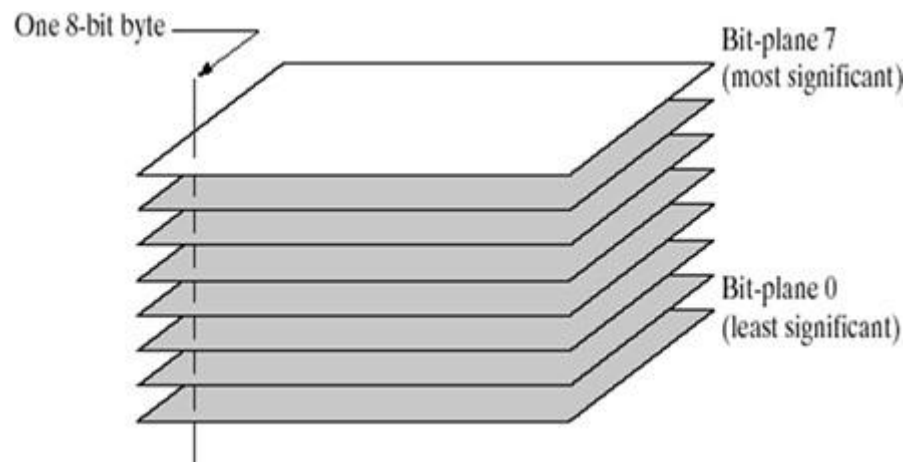


The second approach is to brighten the desired range of gray levels but preserve the background and gray-level tonalities in the image:



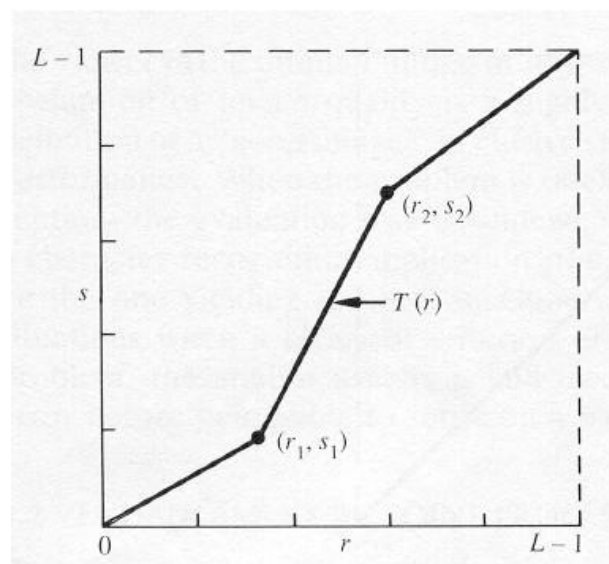
Bit plane slicing

Bit plane slicing is used to highlight the contribution made to the total image appearance by specific bits. Assuming that each pixel is represented by 8 bits, the image is composed of 8 1-bit planes. Plane 0 contains the least significant bit and plane 7 contains the most significant bit. Only the higher order bits (top four) contain visually significant data. The other bit planes contribute the more subtle details. Plane 7 corresponds exactly with an image thresholded at gray level 128.



Contrast stretching

Contrast stretching (often called normalization) is a simple image enhancement technique that attempts to improve the contrast in an image by 'stretching' the range of intensity values it contains to span a desired range of values, *e.g.* the the full range of pixel values that the image type concerned allows. It differs from the more sophisticated histogram equalization in that it can only apply a *linear* scaling function to the image pixel values. As a result the 'enhancement' is less harsh.



The locations of (r_1, s_1) and (r_2, s_2) control the shape of the transformation function.

- If $r_1 = s_1$ and $r_2 = s_2$ the transformation is a linear function and produces no changes.

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- If $r_1=r_2$, $s_1=0$ and $s_2=L-1$, the transformation becomes a thresholding function that creates a binary image.
- Intermediate values of (r_1, s_1) and (r_2, s_2) produce various degrees of spread in the gray levels of the output image, thus affecting its contrast.

Generally, $r_1 \leq r_2$ and $s_1 \leq s_2$ is assumed.

Implementation steps with screenshots:

1. Image Negative

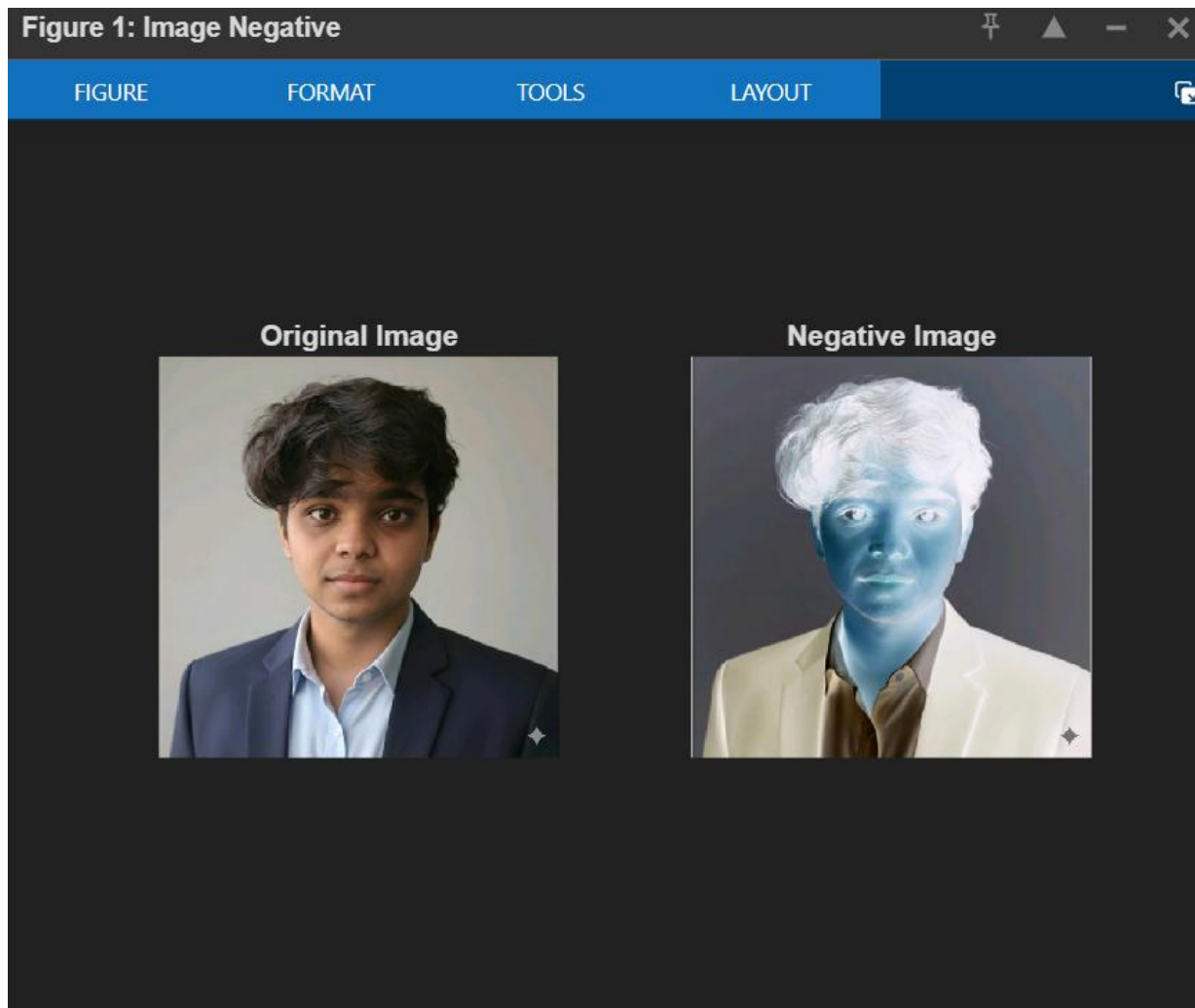
```
clc; clear; close all;

% Read Image
I = imread('IMG_6232.jpg');
I = im2uint8(I);

%% 1. Image Negative
neg = 255 - I;

figure('Name','Image Negative');
subplot(1,2,1);
imshow(I);
title('Original Image');

subplot(1,2,2);
imshow(neg);
title('Negative Image');
```



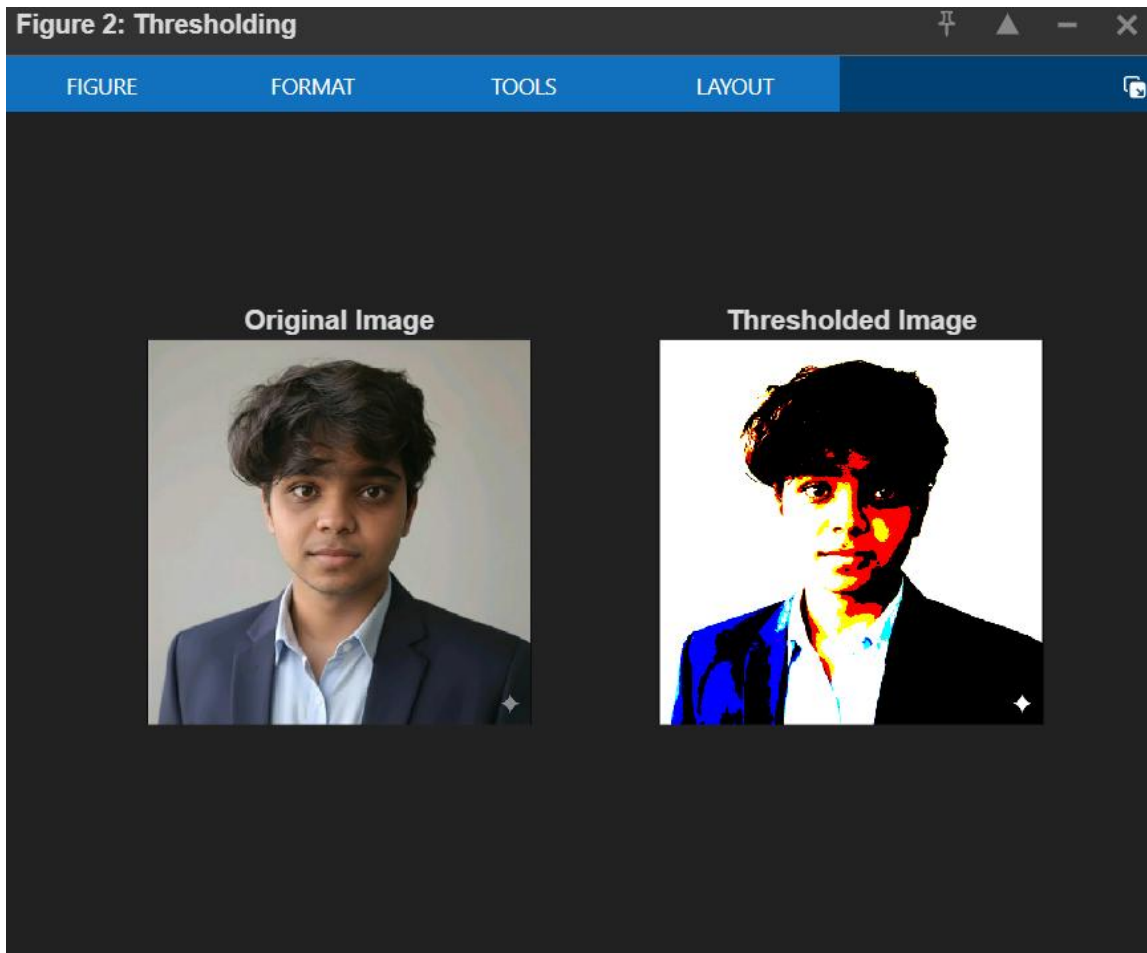
2. Thresholding

```
%% 2. Thresholding
T = 100;
threshold_img = I >= T;
```

```
threshold_img = uint8(threshold_img * 255);
```

```
figure('Name', 'Thresholding');  
subplot(1,2,1);  
imshow(I);  
title('Original Image');
```

```
subplot(1,2,2);  
imshow(threshold_img);  
title('Thresholded Image');
```



3. Gray Scaling

```
%% 3. Gray Level Slicing WITHOUT Background
r1 = 80;
r2 = 160;

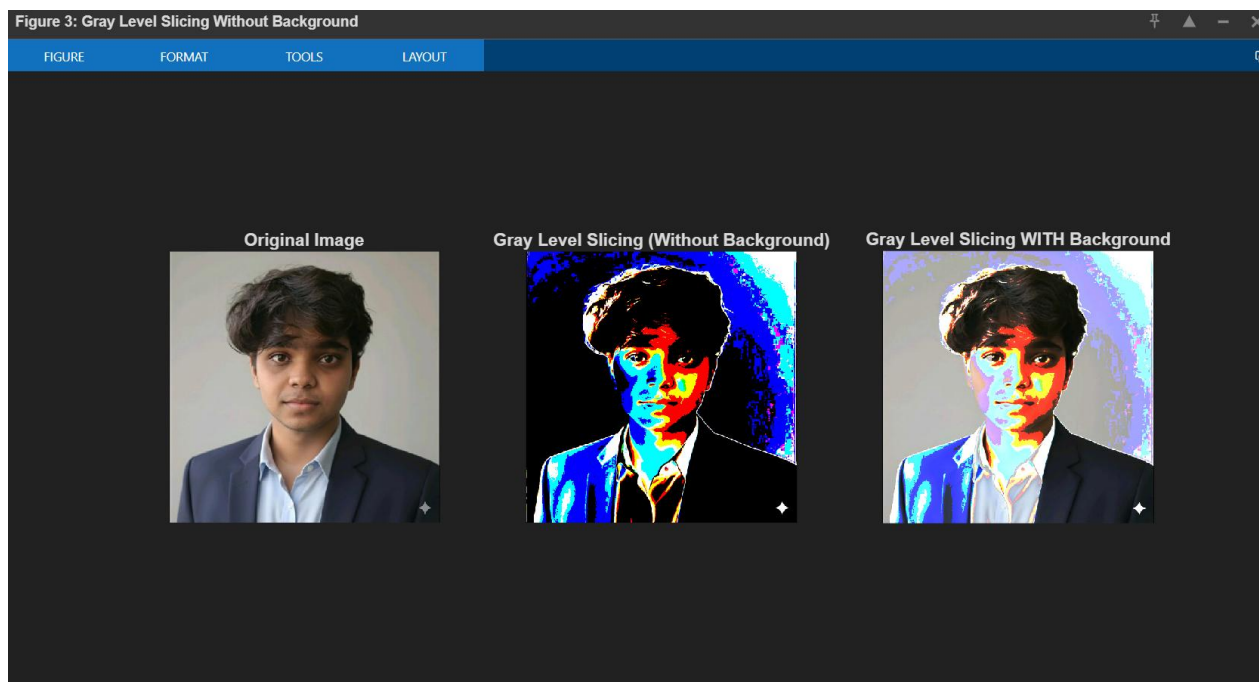
gls_no_bg = zeros(size(I), 'uint8');
gls_no_bg(I >= r1 & I <= r2) = 255;

figure('Name','Gray Level Slicing Without Background');
subplot(1,3,1);
imshow(I);
title('Original Image');

subplot(1,3,2);
imshow(gls_no_bg);
title('Gray Level Slicing (Without Background)');

gls_with_bg = I;
gls_with_bg(I >= r1 & I <= r2) = 255;

subplot(1,3,3);
imshow(gls_with_bg);
title('Gray Level Slicing WITH Background');
```



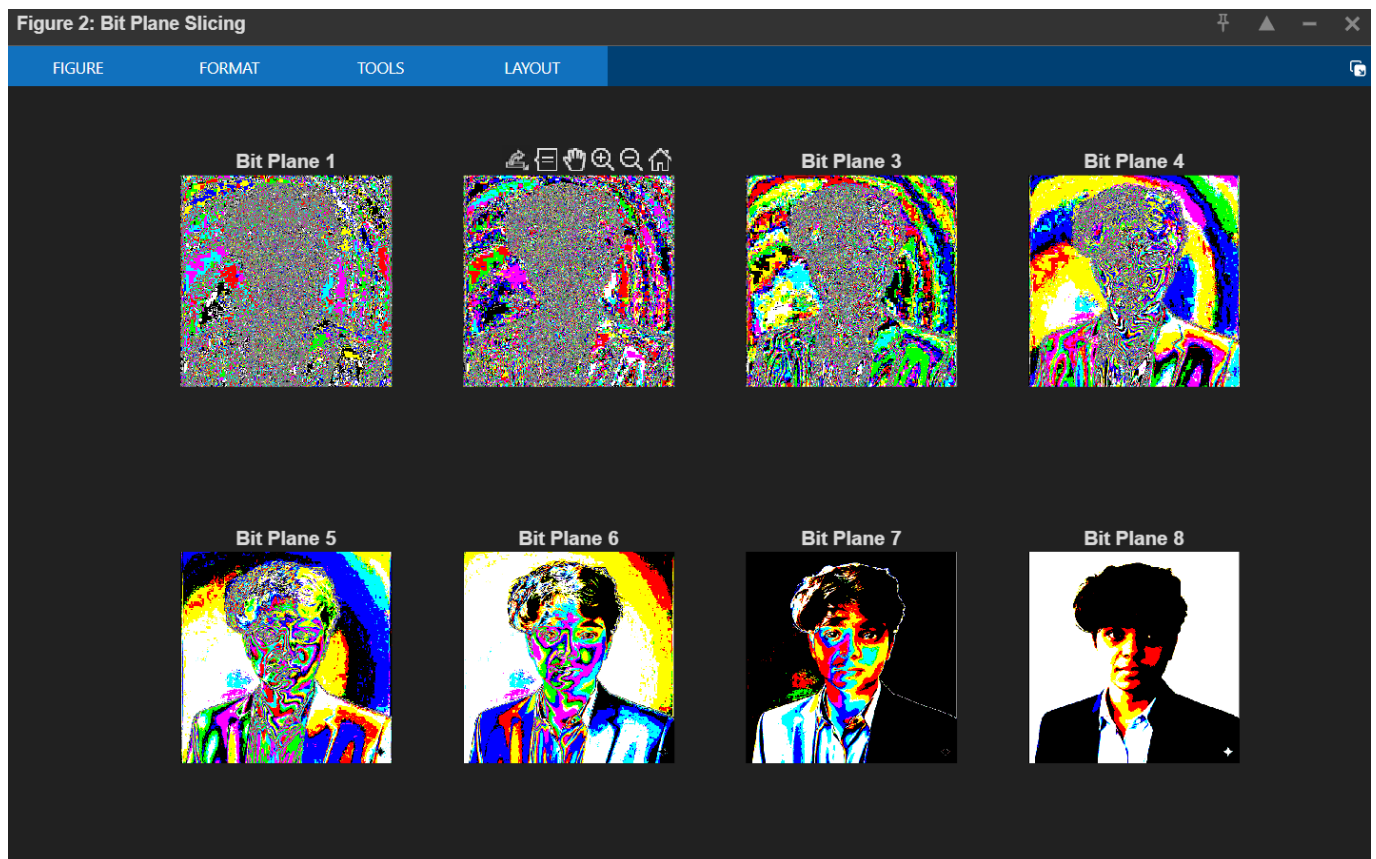
4. Bit Plane Slicing

```

%% 4. Bit Plane Slicing
figure('Name','Bit Plane Slicing');

for k = 1:8
    bit_plane = bitget(I, k);    % Extract kth bit
    bit_plane = uint8(bit_plane) * 255; % Scale for display

    subplot(2,4,k);
    imshow(bit_plane);
    title(['Bit Plane ', num2str(k)]);
end
  
```



5. Contrast Stretching

```
%% 5. Contrast Stretching

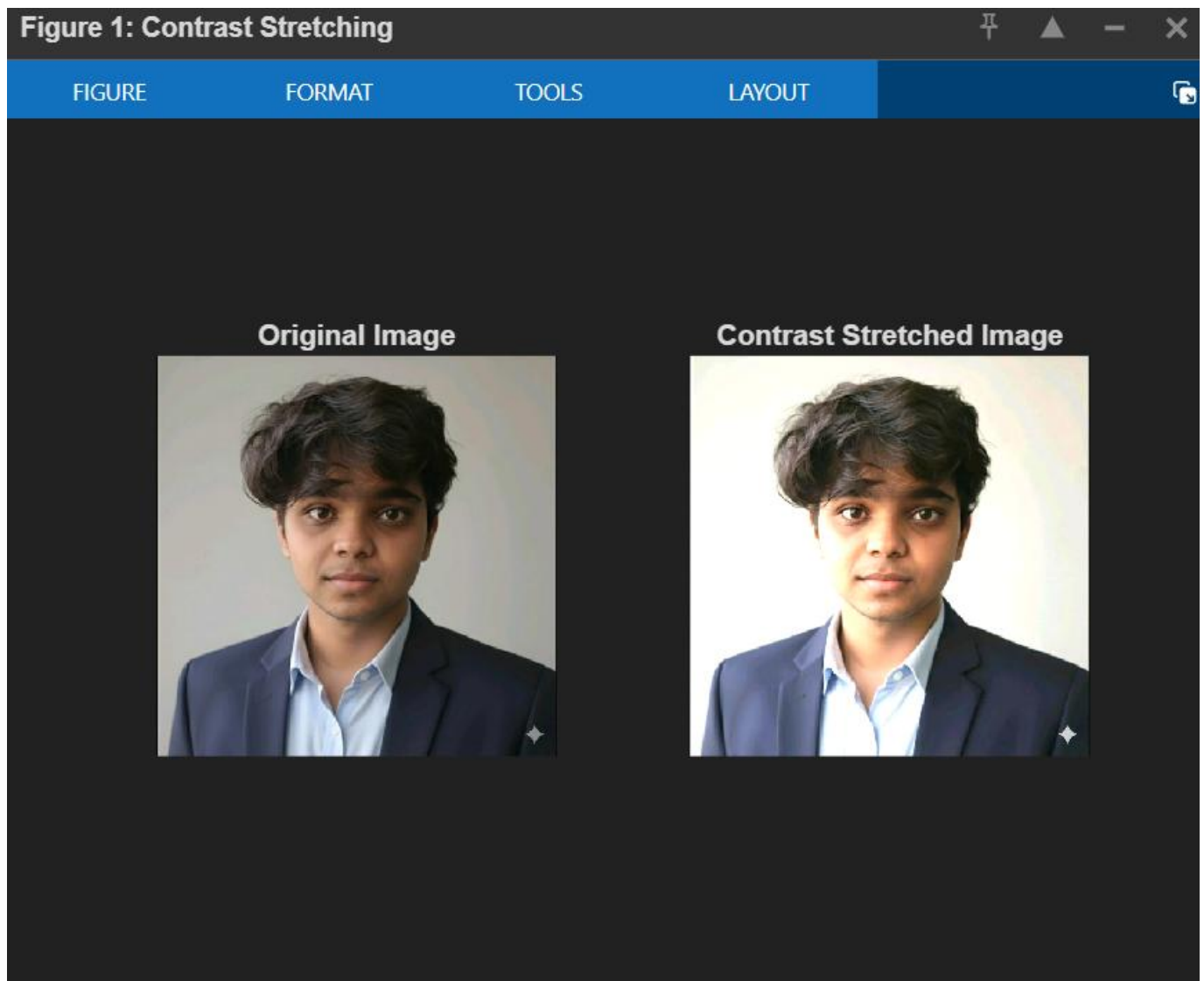
k = 1.5;

contrast_stretch = uint8(double(I) * k);

contrast_stretch(contrast_stretch > 255) = 255;

figure('Name','Contrast Stretching');
subplot(1,2,1);
imshow(I);
title('Original Image');

subplot(1,2,2);
imshow(contrast_stretch);
title('Contrast Stretched Image')
```



Conclusion:-

The experiment successfully demonstrated spatial domain point processing techniques such as image negative, thresholding, gray level slicing, bit plane slicing, and contrast stretching using MATLAB. These techniques enhance image features and improve visual quality.

Date: _____

Signature of faculty in-charge

Post Lab Descriptive Questions

1. Explain the role of bit plane slicing in achieving Steganography concept.

Bit plane slicing decomposes a grayscale image into its individual binary planes, ranging from the **Most Significant Bit (MSB)** to the **Least Significant Bit (LSB)**. In steganography, the **LSB planes** are commonly used to hide secret information because changes in these lower-order bits cause **minimal visual distortion** to the image.

By embedding secret data into the LSBs of the cover image, the hidden information remains imperceptible to the human eye while still being recoverable during extraction. Thus, bit plane slicing enables secure and invisible data hiding, forming the basis of image steganography techniques.

2. Explain the use of gray level slicing

Gray level slicing is used to **highlight a specific range of intensity values** in an image while suppressing or retaining other gray levels. This technique is especially useful when certain objects or regions of interest fall within a known intensity range.

Gray level slicing can be performed **with background**, where non-selected gray levels remain unchanged, or **without background**, where only the selected range is emphasized and the rest is suppressed. It is widely used in medical imaging, satellite image analysis, and defect detection to enhance important features.

Post Lab Objective Questions

1. Thresholding function in contrast stretching creates
 - a) **binary image**
 - b) high quality image
 - c) enhanced image
 - d) low quality image

2. When is the contrast stretching transformation a linear function, for r and s as gray-value of image before and after processing respectively?
 - a) **$r1 = s1$ and $r2 = s2$**
 - b) $r1 = r2, s1 = 0$ and $s2 = L - 1$, L is the max gray value allowed
 - c) $r1 = 1$ and $r2 = 0$
 - d) None of the mentioned

3. Which gray-level transformation increase the dynamic range of gray-level in the image?
 - a) Power-law transformations
 - b) Negative transformations
 - c) **Contrast stretching**
 - d) None of the mentioned

4. When is the contrast stretching transformation a thresholding function, for r and s as gray-value of image before and after processing respectively?
 - a) $r1 = s1$ and $r2 = s2$
 - b) **$r1 = r2, s1 = 0$ and $s2 = L - 1$, L is the max gray value allowed**
 - c) $r1 = 1$ and $r2 = 0$
 - d) None of the mentioned

5. What condition prevents the intensity artifacts to be created while processing with contrast stretching, if r and s are gray-values of image before and after processing respectively?

- a) $r_1 = s_1$ and $r_2 = s_2$
- b) $r_1 = r_2$, $s_1 = 0$ and $s_2 = L - 1$, L is the max gray value allowed
- c) $r_1 = 1$ and $r_2 = 0$
- d) $r_1 \leq r_2$ and $s_1 \leq s_2$**